Joshua Lim

Ph.D. Candidate, Department of Statistical Science, Duke University

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EDUCATION

Duke University **Durham, NC**
Ph.D., Statistical Science expected May 2028
- Advisors: Jerome P. Reiter, Fan Li
- Certificate in College Teaching

University of California, Los Angeles **Los Angeles, CA**
B.S., Applied Mathematics, Statistics and Data Science June 2024
- Summa Cum Laude; GPA 4.00/4.00

HONORS AND AWARDS

Selected Participant, FutureBAProf Workshop August 2026
Tippie College of Business, University of Iowa. One of 22 participants selected from 117 applicants.

Dean's Graduate Fellowship, Duke University 2024
Awarded to incoming Ph.D. students with a record of excellent leadership.

Summa Cum Laude, UCLA 2024
Awarded for a cumulative GPA above 3.985; Dean's Honor List, Fall 2020–Spring 2024.

RESEARCH INTERESTS

Causal inference, heterogeneous treatment effects, missing data, machine learning methods for causal estimation, statistical computing and software, biostatistics.

WORKING PAPERS AND MANUSCRIPTS IN PREPARATION

1. **Lim, J.** and Reiter, J. P. "Bayesian Treatment Imputation for CATE Estimation Under Missing Binary Treatments." In preparation.
   - Proposes a data augmentation Gibbs sampler that jointly updates a BART outcome model, a probit propensity model, and the missing treatment indicators in a single chain, propagating the parameter uncertainty that sequential imputation discards.
2. **Lim, J.** and Li, F. "Bayesian Causal Inference: A Practical Guide." In preparation for submission to the *Journal of Statistical Software*.
   - Introduces the `BayesCausal` R package: Bayesian estimation of average and heterogeneous treatment effects under unconfounded, instrumental variable, and panel designs, behind a single formula interface with Stan HMC posterior inference.

PEER-REVIEWED PUBLICATIONS

1. Zhu, D., Kranz, P. G., Edelmuth, D. G. L., **Lim, J.**, Christensen, S., Madhavan, A. A., & Amrhein, T. J. (2025). Maximizing the conspicuity of CSF–venous fistulas on CT myelography: Assessment of contrast density and timing effects. *American Journal of Neuroradiology*. doi:10.3174/ajnr.A8972

Conference Abstracts and Presentations

1. Heterogeneous treatment effect estimation with missing data. *FutureBAProf Workshop*, Tippie College of Business, University of Iowa. August 2026. (Invited 3-minute research pitch.)
2. Maximizing the conspicuity of CSF–venous fistulas on CT myelography: Assessment of contrast density and timing effects. *Radiological Society of North America Annual Meeting*. December 2025.
3. Enhancing conspicuity of CSF–venous fistulas on CT myelography: Assessment of contrast density and timing effects. *Intracranial Hypotension Conference*. June 2025.
4. Tirumalasetty, J., **Lim, J.**, Eggert, L., Arroyo, A., & Nassikas, N. (2025). Associations of air pollution and heat with asthma healthcare visits in California counties — an ecological study. *AAAAI/WAO Joint Congress*; published in *Journal of Allergy and Clinical Immunology*, 155(2 Suppl.), AB309. doi:10.1016/j.jaci.2024.12.951
5. **Lim, J.**, Cao, S., Chinthrajah, S., & Sindher, S. (2023). The impact of persistent gastrointestinal symptoms on completion of oral immunotherapy in clinical trials. *AAAAI Annual Meeting*; published in *Journal of Allergy and Clinical Immunology*, 151(2). doi:10.1016/j.jaci.2022.12.110 Also presented at *GA^{2}LEN ANACARE GAFA*, September 2022.

Software

- **BayesCausal** — R package for Bayesian causal inference under unconfounded, instrumental variable, and panel designs; formula interface, structured priors, and Stan HMC posterior inference.

Research Experience

Duke University, Department of Statistical Science **Durham, NC**
Graduate Research Assistant May 2025 – Present

- Developing Bayesian methods for estimating heterogeneous treatment effects when the treatment indicator is partially missing, combining BART-based outcome models with a Gibbs sampler that imputes the missing treatment jointly with the response surface.
- Building a Bayesian implementation of several causal inference methods, with an accompanying R package.
- Integrating large language models and generative AI into the research workflow — developing and scaling simulation studies for benchmarking and method testing, and managing high-performance computing jobs on the cluster.
- Advised by Professors Jerome Reiter and Fan Li.

Qing Zhou Research Group, UCLA **Los Angeles, CA**
Undergraduate Researcher February 2023 – June 2024

- Researched structure learning of causal DAGs under Professor Qing Zhou.
- Extended the multi-neighborhood learning algorithm to discrete data by adding conditional independence tests and related procedures.
- Validated the algorithm through simulation studies on the Hoffman2 cluster using datasets generated from bnlearn repository parameters.
- Implemented in R, Rcpp, and RcppArmadillo.

Statistical Consulting and Applied Experience

UC San Diego School of Medicine, Moores Cancer Center **La Jolla, CA**
Biostatistician December 2025 – Present

- Modeling treatment-related complications between patients receiving immunotherapy with radiation versus concurrent chemoradiation with cisplatin.

Duke University School of Medicine **Durham, NC**
Biostatistician May 2025 – Present
- Assessed the relative impact of contrast density and image acquisition timing on CSF–venous fistula visibility; results published in *AJNR*.
- Analyzed incomplete survey data on patient book preference to inform pediatric practice.
- Authored abstracts, presentations, and manuscript submissions for multiple medical conferences.

Stanford School of Medicine, Pulmonary, Allergy and Critical Care Medicine **Stanford, CA**
Biostatistician and Data Manager June 2024 – March 2025
- Identified environmental factors associated with increased asthma and COPD emergency visits and hospitalizations, tracking regional and county-level differences with attention to vulnerable populations.
- Second author on the abstract and presentation submitted to the 2025 AAAAI/WAO Joint Congress.

Stanford School of Medicine, Sean N. Parker Center for Allergy and Asthma Research **Stanford, CA**
Data Analyst June 2022 – May 2023
- Analyzed chronic GI symptoms among patients in oral immunotherapy clinical trials; first author on the resulting AAAAI abstract and poster.
- Built machine learning models to automatically map CTCAE, Bock, and CoFAR 1 allergic reaction grades onto the CoFAR 3 scale.
- Developed an interactive R Shiny application displaying patient enrollment, demographic, and forecast data, used by researchers to prepare NIH summaries.

TEACHING EXPERIENCE

Duke University, Department of Statistical Science **Durham, NC**
Instructor of Record — STA 199L: Introduction to Data Science and Statistical Thinking Summer 2026
- Sole instructor for a for-credit core statistics course: designed and delivered 29 lectures, six labs, and two exams.
- Received a unanimous 5.0/5.0 on student instructor evaluations.

Duke University, Department of Statistical Science **Durham, NC**
Instructor, Linear Algebra Bootcamp for Incoming Ph.D. Students August 2026
- Taught a one-day linear algebra review preparing incoming Ph.D. students for STA 721L: Linear Models
- Authored a 69-slide deck with 21 in-class practice problems and a companion worked-solutions manual.

Johnson C. Smith University **Charlotte, NC**
Instructor, Introduction to Statistics for Health Science Using R May 2026
- Designed and taught an NSF-funded, one-day workshop introducing statistics and R to 12 undergraduates with no prior programming experience, built around an NHANES public-health case study.
- Wrote six slide modules and paired instructor/student scripts in base R, so students needed no package installation to follow along.

Duke University, Department of Statistical Science **Durham, NC**
Teaching Assistant — STA 199CCL: Introduction to Data Science and Statistical Thinking Fall 2026

Duke University, Department of Statistical Science **Durham, NC**
Teaching Assistant, M.S.S. Bootcamp for Incoming Master's Students Summer 2026

UCLA, Department of Statistics and Data Science **Los Angeles, CA**
Teaching Assistant January 2024 – June 2024

- Stats 13: Introduction to Statistical Methods for Life and Health Sciences; Stats 102B: Computation and Optimization for Statistics.
- Wrote original homework assignments and practice exams, led office hours and review sessions, and graded for sections totaling over 180 students.

Math Speeder Upper Course (MSUC) **Remote**
Instructor and Curriculum Designer June 2021 – August 2021
- Built a free, fully asynchronous introductory multivariable calculus course — 22 lectures, assignments, quizzes, two midterms, three projects, and a final — designed to be accessible regardless of student background. Still in use by new students each year.

Earlier teaching (2019–2021). Teaching Assistant, Menlo School (AP Statistics, AP Calculus BC), Atherton, CA; instructor and curriculum designer, Physics Catcher Upper Course; private tutor in mathematics, statistics, and physics.

LEADERSHIP AND SERVICE

Duke University, Department of Statistical Science **Durham, NC**
Mentor, Introduction to Undergraduate Research Workshop May 2025
- Mentored a team of six undergraduates through an applied statistical analysis of perceived quality of life in Durham, NC.
- Ran mini-workshops on research practice, coding practice, and collaboration for all 20+ program participants.
- Completed mentor training under Dr. Joan Combs Durso.

Duke University, Department of Statistical Science **Durham, NC**
Member, Graduate Consultative Committee August 2024 – Present
- Work with the Director of Graduate Studies and department chair on academic and personal needs of more than 50 Ph.D. students.
- Plan and run admitted-student open house events, and organize department social events including the Halloween and winter parties and the intramural volleyball team.

UCLA Club Gymnastics **Los Angeles, CA**
President August 2022 – June 2024
- Led a nationally ranked club team of over 60 members: organized meetings, home competitions, and travel meets; handled governing-body communication, competition registration, and fundraising.

TECHNICAL SKILLS

- **Languages:** R, Python, SQL, C++, MATLAB, Java
- **Statistical computing:** Rcpp/RcppArmadillo, Stan, BART, tidyverse, Shiny
- **Tools:** Git, LaTeX, Quarto, high-performance computing clusters (SLURM), REDCap, large language model and generative AI tools